
Symmetries in Physics — Exercise Sheet 11

SO(3), the double cover, and spherical harmonics

Variant: Questions only

Exercise 1 Prove the Euler decomposition: every $R \in \text{SO}(3)$ can be written $R = R_z(\alpha)R_y(\beta)R_z(\gamma)$. Identify where the decomposition fails to be unique.

Exercise 2 For $U = \cos \frac{\theta}{2} \mathbb{1} - i \sin \frac{\theta}{2} \hat{n} \cdot \vec{\sigma} \in \text{SU}(2)$, compute $\Phi(U)$ acting on $\vec{x}$ via $U(\vec{x} \cdot \vec{\sigma})U^\dagger$, and verify both the Rodrigues formula and the two-to-one property.

Exercise 3 Let P_ℓ be the homogeneous polynomials of degree ℓ on $\mathbb{R}^3$. (i) Show $\dim P_\ell = \binom{\ell+2}{2}$. (ii) Using the Fischer pairing, in which $\langle r^2 f, g \rangle = \langle f, \Delta g \rangle$, prove that $\Delta: P_\ell \rightarrow P_{\ell-2}$ is surjective, and deduce $\dim H_\ell = 2\ell + 1$ for the harmonic subspace.

Exercise 4 Starting from the highest-weight vector $w_2 = (x + iy)^2$, generate the five harmonics of $\ell = 2$ by repeated application of $L_- = -z(\partial_x - i\partial_y) + (x - iy)\partial_z$, and check the chain terminates.

Exercise 5 Prove directly — following the loop $R_z(\theta)$, $\theta \in [0, 2\pi]$ — that a continuous representation of $\text{SO}(3)$ cannot contain half-integer spin.

Exercise 6 A particle on the unit sphere has $H_0 = L^2/2I$. (i) State the spectrum and degeneracies. (ii) A weak field adds $V = \varepsilon \cos^2 \theta$, breaking $\text{SO}(3)$ to rotations about $\hat{z}$ (and reflections). Compute the first-order splitting of the $\ell = 1$ level and exhibit the residual degeneracy.

Exercise 7 The spin- j Wigner matrix is $D_{m'm}^j(\alpha, \beta, \gamma) = e^{-im'\alpha} d_{m'm}^j(\beta) e^{-im\gamma}$, with the reduced matrix $d^j(\beta) = e^{-i\beta J_y}$ in the spin- j representation. (i) For $j = 1$, in the basis $m = 1, 0, -1$, compute $d^1(\beta)$ explicitly. (ii) Read off the Euler-angle phase structure of D^1 and verify that D^1 is unitary.